

Survey Questionnaire on Decision-Making Processes and Influencing Factors of Apple Growers

In recent years, with the evolving fruit market and the increasing complexity of decision-making in orchard management, issues such as pesticide misuse and inappropriate treatments have emerged, leading to higher production costs. Additionally, limited access to market information has placed apple growers at a disadvantage in trading activities. This questionnaire is designed to investigate these processes. Through analysis of the responses, we aim to identify ways to reduce operational costs, minimize unnecessary expenditures, and improve growers' profitability.

Background Information

1. Name [Open-ended]

2. Gender [Single choice]

- ☐ Male
- ☐ Female

3. Age [Open-ended]

4. Educational Level [Single choice]

- ☐ Primary school diploma
- ☐ Junior high school diploma
- ☐ High school diploma
- ☐ Associate degree
- ☐ Bachelor's degree

- Master's degree or above

5. Postal Code [Open-ended]

6. Contact Information [Open-ended]

Mobile number

7. Location of the Orchard (Precise to Province, City, District, and Village) [Open-ended]

8. Orchard Area (1 mu is equal to 1/15 of a hectare) [Single choice]

- 0–5 Mu
- 5–10 Mu
- 10–20 Mu
- 20–30 Mu
- 30–50 Mu
- 50–100 Mu
- 100–200 Mu
- 200–500 Mu
- 500–800 Mu
- 800–1000 Mu
- 1000–2000 Mu
- 2000 Mu or more

9. Number of Family Labor Forces [Single choice]

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ >3

10. Have you participated in professional agricultural skills training? [Single choice]

- ☐ Yes
- ☐ No

11. Main Cultivated Varieties [Multiple choice]

- ☐ Fuji
- ☐ Qin Crops
- ☐ Qinyang
- ☐ Yanfu
- ☐ Gala
- ☐ Red General
- ☐ Yan Gala
- ☐ Qin Crown
- ☐ Changfu
- ☐ Golden Delicious
- ☐ Other

12. Age of Apple Trees [Single choice]

- ☐ 1–3
- ☐ 3–6
- ☐ 6–9
- ☐ >10

13. Orchard Topography [Single choice]

- ☐ Plain
- ☐ Mountainous
- ☐ Steep slope
- ☐ Hilly
- ☐ Along the Highway
- ☐ Other

14. Profile of Orchard Management Personnel [Single choice]

- ☐ Orchard Owner Manages the Orchard Part-time
- ☐ Orchard Owner Manages the Orchard Full-time
- ☐ Managed by hired manager/agent
- ☐ Corporate base under a company system

15. Have you participated in a farmers cooperative? [Single choice]

- ☐ Previously participated, not currently participating
- ☐ Have not participated
- ☐ Already participated

Planting Practices

16. Cultivation Model [Single choice]

- ☐ Dwarfing and dense planting
- ☐ Standard tree and sparse planting
- ☐ Other

17. What is the information source for cultivation pattern decision-making? [Multiple choice]

- ☐ Local experts
- ☐ Farmer peers
- ☐ Personal experience
- ☐ Mass media
- ☐ Agricultural Technology Extension Service Centers
- ☐ Farmer cooperatives
- ☐ Agri-chemical Dealers

18. Have you modified the original cultivation pattern? [Single choice]

- ☐ Yes
- ☐ No

19. What is the revised cultivation pattern? [Single choice] (If Q18 = Yes)

- ☐ Standard tree and sparse planting

- Dwarfing and dense planting
- Other

20. What is the information source for the decision to change the cultivation pattern? [Multiple choice] (*If Q19 = any option*)

- ☐ Local experts
- ☐ Farmer peers
- ☐ Personal experience
- ☐ Mass media
- ☐ Agricultural Technology Extension Service Centers
- ☐ Farmer cooperatives
- ☐ Agri-chemical Dealers

21. What type of seedlings was selected? [Single choice]

- Virus-Free Seedlings
- Common seedlings
- Mixed use

22. What is the information source for the seedling selection decision? [Multiple choice]

- ☐ Local experts
- ☐ Farmer peers
- ☐ Personal experience
- ☐ Mass media
- ☐ Agricultural Technology Extension Service Centers

- ☐ Farmer cooperatives
- ☐ Agri-chemical Dealers

23. What are your priority considerations when selecting a variety? [Single choice]

- ☐ Price
- ☐ Quality
- ☐ New Varieties
- ☐ Following the majority's choice
- ☐ Innovative Promotion
- ☐ Varieties specified by buyers or merchants
- ☐ Random Experimentation
- ☐ Farmer cooperatives
- ☐ Agricultural Technology Extension Services

24. What are the priority considerations in decision-making for planting time? [Single choice]

- ☐ Based on personal convenience
- ☐ Personal Experiences
- ☐ Following the majority's choice
- ☐ Local experts
- ☐ Guidance from seedling sellers

25. What is the number of seedlings planted per mu? (1 mu is equal to 1/15 of a hectare) [Open-ended]

26. What are the acquisition channels for the seedlings? [Multiple choice]

- ☐ Purchased from seedling enterprises
- ☐ Provided by the local government
- ☐ Purchased from the seedling market
- ☐ Purchased from local seedling farmers
- ☐ Self-propagated and self-cultivated
- ☐ From friends' orchards

27. Do the seedlings conform to the product description? [Single choice]

- ☐ Compliant
- ☐ Non-compliant

28. What treatments were carried out before cultivation? [Multiple choice]

- ☐ Soil sterilization
- ☐ Soil insecticide application
- ☐ Seedling disinfection
- ☐ None

29. What is the information source for the decision-making on pre-planting disinfection treatment? [Multiple choice] *(If Q28 includes any of the first three options)*

- ☐ Local experts
- ☐ Farmer peers
- ☐ Personal experience

- ☐ Mass media
 - ☐ Agricultural Technology Extension Service Centers
 - ☐ Farmer cooperatives
 - ☐ Technical advice from seedling sellers
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Management Practices

30. The main diseases occurring in orchards [Multiple choice]

- ☐ Apple rust disease
- ☐ Canker disease
- ☐ Powdery mildew disease
- ☐ Rust fruit disease
- ☐ Apple Botryosphaeria dothidea
- ☐ Apple Alternaria blotch
- ☐ Apple scab (*Venturia inaequalis*)
- ☐ Apple anthracnose
- ☐ Other

31. The main pests occurring in orchards [Multiple choice]

- ☐ Spider Mite
- ☐ Aphid
- ☐ Codling Moth
- ☐ Borer
- ☐ Scale insects

- ☐ Longhorn Beetle
- ☐ Anomala cuprea
- ☐ Other

32. The Level of Awareness of Pest and Disease Control [Single choice]

- ☐ Behavior of Preventive Fungicide Application
- ☐ Lacking Preventive Fungicide Application Behavior, Only Treating Pests and Diseases Once They Manifest
- ☐ Possessing the Awareness of Preventive Fungicide Application but Lacking the Time to Implement It
- ☐ Possessing the Awareness of Preventive Fungicide Application but Lacking Economic Capacity
- ☐ Following Others Pest and Disease Control Strategies

33. What type of plant protection agents are chosen? [Multiple choice]

- ☐ Chemical
- ☐ Physical
- ☐ Biological
- ☐ Uncertain of the type

34. How many times are fungicides used in orchards each year? [Open-ended]

35. What are the types of fungicide ingredients used? [Multiple choice]

- ☐ Plant-derived
- ☐ Organic

- ☐ Inorganic
- ☐ Microbial
- ☐ Uncertain of the type

36. What are the acquisition channels for the fungicides? [Multiple choice]

- ☐ Agri-chemical Dealers
- ☐ Large-scale farmer
- ☐ Farmer Cooperatives
- ☐ Leading enterprise
- ☐ Online shopping
- ☐ Farmer Peers
- ☐ Provided by the local government

37. What is the information source for the decision-making on fungicide selection? [Multiple choice]

- ☐ Local experts
- ☐ Farmer peers
- ☐ Personal experience
- ☐ Mass media
- ☐ Agricultural Technology Extension Service Centers
- ☐ Farmer cooperatives
- ☐ Recommendations from agricultural supply stores

38. Whether to repurchase fungicides due to poor efficacy? [Single choice]

- ☐ Yes

- No

39. What are the sources of decision-making information when repurchasing fungicides? [Multiple choice]

(Dependent on Option 1 of Question 38)

- ☐ Local experts ☐ Farmer peers ☐ Personal experience
- ☐ Mass media ☐ Agricultural Technology Extension Service Centers
- ☐ Farmer cooperatives ☐ Agri-chemical Dealers

40. What is the information source for the decision-making on fungicide application strategy? [Multiple choice]

- ☐ Farmer peers ☐ Agri-chemical Dealers
- ☐ Product Manual ☐ Local Experts
- ☐ Mass Media ☐ Personal Experiences

41. In your decision-making process, what is the order of influence of the following sources of information? [Ranking question, please fill in numbers in the brackets]

- [] Farmer peers
- [] Agri-chemical Dealers
- [] Product Manual
- [] Expert advice
- [] Mass Media
- [] Personal Experiences

42. What are the primary considerations when selecting fungicides? [Single choice]

- ☐ Price ☐ Effectiveness ☐ Brand
- ☐ Promotion ☐ Previously used by friends ☐ Local Experts
- ☐ Currently undergoing a promotional campaign

43. What are the priority considerations when formulating a fungicide spraying plan? [Single choice]

- ☐ To take preventive measures in advance
- ☐ Personal available time
- ☐ Weather factors
- ☐ Whether the disease has already occurred
- ☐ Product Manual
- ☐ Following other farmers
- ☐ Knowledge from agricultural technology training

44. Are you aware of when to apply what fungicides? [Single choice]

- ☐ Yes
- ☐ No

45. Where did you acquire the knowledge regarding the aforementioned issues? [Multiple choice]

- ☐ Local experts ☐ Farmer peers ☐ Personal experience
- ☐ Mass media ☐ Agricultural Technology Extension Service Centers
- ☐ Farmer cooperatives ☐ Agri-chemical Dealers

46. Which equipment do you use for the application of fungicides? [Multiple choice]

- ☐ Backpack-style manual sprayer ☐ Unmanned Aerial Vehicle (UAV)
- ☐ Small mobile sprayer ☐ Hand-pushed gasoline-powered sprayer
- ☐ Electric sprayer ☐ Other

47. Describe your level of satisfaction with the current orchard pest and disease control effectiveness [Single choice]

- ☐ Very Dissatisfied
- ☐ Dissatisfied
- ☐ General
- ☐ Satisfied
- ☐ Very Satisfied

48. What is the average cost of fungicides used per mu of land? (The unit is: Yuan) [Single choice]

(Unit: RMB per year)

- ☐ 0–100 ☐ 100–200 ☐ 200–300 ☐ 300–400
- ☐ 400–500 ☐ 500–600 ☐ 600–700 ☐ 700–800
- ☐ 800–900 ☐ 900–1000 ☐ >1000

49. What type of fertilizer is applied in the orchard? [Multiple choice]

- ☐ Farmyard Manure ☐ Commercial Organic Fertilizer
- ☐ Biofertilizer ☐ Long-acting Compound Fertilizer
- ☐ Humic Acid-based Compound Micro-fertilizer ☐ Chemical fertilizer

- ☐ Blended Fertilizer ☐ Other

50. What is the information source for the decision-making on fertilizer selection? [Multiple choice]

- ☐ Local experts ☐ Farmer peers ☐ Personal experience
- ☐ Mass media ☐ Agricultural Technology Extension Service Centers
- ☐ Farmer cooperatives ☐ Agri-chemical Dealers

51. Have You Changed Your Fertilizer Purchase Decision Due to Unsatisfactory Results? [Single choice]

- ☐ Yes
- ☐ No

52. Information Sources for Re-selection Decisions [Multiple choice]

(Dependent on Option 1 of Question 51)

- ☐ Local experts ☐ Farmer peers ☐ Personal experience
- ☐ Mass media ☐ Agricultural Technology Extension Service Centers
- ☐ Farmer cooperatives ☐ Agri-chemical Dealers

53. Fertilizer Purchase Channels [Multiple choice]

- ☐ Agri-chemical Dealers ☐ Large-scale growers ☐ Cooperatives
- ☐ Leading enterprises ☐ Online purchases ☐ Acquaintances
- ☐ Government subsidies

54. What are your priority considerations when selecting fertilizers? [Single choice]

- ☐ Price ☐ Effectiveness ☐ Ease of use ☐ Promotional activities
- ☐ Repeat purchase ☐ Brand effect ☐ Friends have used it ☐ Other

55. What are the priority considerations for the fertilizer application strategy? [Single choice]

- ☐ Phenological patterns ☐ Product Manual
- ☐ Apply fertilizers only when deficiency symptoms appear ☐ The strategy of following friends' approaches
- ☐ Minimize application to save costs ☐ Use fertilizers whenever they are available

56. How many times are fertilizers applied each year? [Open-ended]

57. What is the average amount of fertilizer applied per mu? (1 jin = 0.5 kilograms) [Open-ended]

58. What is the cost of fertilizer application per mu? (unit: yuan) [Single choice]

(Unit: RMB/Mu)

- ☐ 0–50 ☐ 50–100
- ☐ 100–200 ☐ 200–300
- ☐ 300–400 ☐ 400–500
- ☐ 500–600 ☐ 600–700

59. What type of fruit bag is used for apple bagging? [Multiple choice]

- ☐ Paper bag ☐ Plastic film bag

- ☐ Paper bag + Plastic film bag ☐ Liquid film bags
- ☐ Other

60. What are the acquisition channels for the fruit bags? [Multiple choice]

- ☐ Agri-chemical Dealers ☐ Large-scale farmers ☐ Leading enterprises
- ☐ Market ☐ Farmer Cooperatives ☐ Online shopping
- ☐ Farmer Peers ☐ Provided by the local government

61. What is the cost of purchasing fruit bags per mu? (unit: yuan) [Single choice]

(Unit: RMB)

- ☐ 0–100 ☐ 100–200 ☐ 200–300
- ☐ 300–400 ☐ 400–500 ☐ 500–600
- ☐ >600

62. Do you hire specialized labor for bagging annually? [Single choice]

- ☐ Yes
- ☐ No

63. What is the Labor cost for bagging and debagging each year? (unit: yuan)
[Single choice]

(Cost per person per day)

- ☐ 0–100
- ☐ 100–200
- ☐ 200–300

- 300–400

(Dependent on Option 1 of Question 62)

64. How many workers are hired for bagging each year? (including bag removal)
[Open-ended]

(Dependent on Option 1 of Question 62)

65. How many days are required for bagging each year? (including bag removal)
[Open-ended]

(Unit: days)

66. What is the average number of bags per acre? [Open-ended]

(Unit: pieces)

67. What is the information source for the decision-making on bagging strategy?
[Multiple choice]

- ☐ Local experts ☐ Farmer peers ☐ Personal experience
- ☐ Mass media ☐ Agricultural Technology Extension Service Centers
- ☐ Farmer cooperatives ☐ Agri-chemical Dealers

68. What are the priority considerations for fruit bag selection? [Single choice]

- Price ○ Promotional activities ○ Promotional activities
- Friends have used it ○ Varieties ○ Operational difficulty
- Fruit growth condition

69. What are the main considerations for selecting the timing of bagging? [Single choice]

- ☐ Weather Factors ☐ Fruit growth condition
- ☐ Growth Stage ☐ Free time
- ☐ The strategy of following friends' approaches ☐ Labor cost

70. What are the primary considerations for selecting the timing of debugging? [Single choice]

- ☐ Tree growth condition ☐ Fruit quality ☐ Weather Factors
- ☐ The strategy of following friends' approaches ☐ Free time
- ☐ Phenological patterns ☐ Labor cost

71. What agricultural machinery is available in the orchard? [Multiple choice]

- ☐ Weed trimmer ☐ Fertilizer spreader ☐ Tractor
- ☐ Backpack-style manual sprayer ☐ Drones ☐ Mounted spray equipment
- ☐ Electric flower and fruit thinning tools ☐ Electric/Pneumatic pruning shears
- ☐ Other

72. What natural disasters have occurred in the orchard? [Multiple choice]

- ☐ Drought ☐ Frost
- ☐ Heavy rain occurs during the flowering period ☐ Hail
- ☐ Other ☐ No disasters occurred

73. Number of Disaster Occurrences in the Past Three Years [Matrix text input]

(Enter a number between 0 and 15 for each type)

Frost: _____

Hail: _____

Heavy rain occurs during the flowering period: _____

Drought: _____

74. What are the channels for receiving disaster warning information? [Multiple choice]

- ☐ Weather forecasting software
- ☐ Broadcasting and television broadcasts
- ☐ Friends' alerts
- ☐ Agricultural Technology Extension Services
- ☐ Referencing historical data
- ☐ Personal Experiences

75. Severity of Different Types of Disasters [Matrix scale question]

(0 = No occurrence, 1 = Mild, 2 = Moderate, 3 = Severe)

Disaster Type	0	1	2	3
Frost	☐	☐	☐	☐
Hail	☐	☐	☐	☐
Heavy rain occurs during the flowering period	☐	☐	☐	☐
Drought	☐	☐	☐	☐

76. What are the information sources for decision-making in disaster prevention measures? [Multiple choice]

- ☐ Local experts
- ☐ Official Guidance

- ☐ Personal Experiences
- ☐ The strategy of following friends' approaches
- ☐ Agricultural Technology Extension Service Centers

77. What are the information sources for decision-making on post-disaster management measures in orchards [Multiple choice]

- ☐ Local experts
- ☐ Official Guidance
- ☐ Personal Experiences
- ☐ The strategy of following friends' approaches
- ☐ Agricultural Technology Extension Service Centers

(Dependent on selection of any disaster in Question 72)

78. How are the losses in the orchard following the disaster? [Single choice]

(0–7, where 0 = No impact, 7 = Total crop failure)

- ☐ 0 - None ☐ 1 - Very mild ☐ 2 - Mild
- ☐ 3 - General ☐ 4 - Less Severe
- ☐ 5 - Severe ☐ 6 - Very Severe ☐ 7 - Complete Crop Loss

(Dependent on selection of any disaster in Question 72)

79. What is the irrigation method used in the orchard? [Multiple choice]

- ☐ Flood irrigation
- ☐ Drip irrigation
- ☐ Subsurface

- ☐ Long Furrow Irrigation
- ☐ Narrow Furrow Irrigation
- ☐ Rainwater harvesting

80. What is the water source for the orchard? [Multiple choice]

- ☐ Natural water sources
- ☐ Tap water
- ☐ Rainwater harvesting
- ☐ Reservoir Water Diversion
- ☐ Other

81. What are the priority considerations in decision-making for irrigation water usage? [Single choice]

- ☐ The strategy of following friends' approaches
- ☐ Presence of drought stress
- ☐ The current growth condition of the fruit trees
- ☐ The growth requirements of fruit trees
- ☐ Weather Factors, for example, has it rained recently?
- ☐ Financial constraints

82. What is the average irrigation volume per mu per application? (Unit: m³)
[Open-ended]

83. What is the irrigation cost per mu? (Unit: Yuan) [Open-ended]

(Includes infrastructure + water fees, unit: RMB)

84. What are the priority considerations when choosing the irrigation timing?
[Single choice]

- The current growth status of the fruit trees
- Degree of drought stress
- Weather Factors
- The strategy of following friends' approaches
- Irrigate when water access is convenient
- The growth requirements of fruit trees

Marketing and Sales Practices

85. Who sets the price when commercial fruit is sold? [Single choice]

- Self-pricing
- Pricing by buyers
- Broker pricing

86. Who establishes the quality grades when commercial fruit is sold? [Single choice]

- Quality grading by farmers
- Quality grading by buyers
- Quality grading by brokers

87. What is the price per jin of commercial fruit? (Note: "jin" is approximately 0.5 kilograms) [Open-ended]

(RMB per jin)

88. What are the earnings from commercial fruit per mu? (Unit: mu) [Open-ended]

(Sales income, in RMB)

89. What are the sales channels? [Multiple choice]

- ☐ Self-production and self-marketing
- ☐ Purchase by fruit merchants
- ☐ Purchase by out-of-town buyers
- ☐ Retailers' acquisition
- ☐ Local fruit company
- ☐ Farmer Cooperatives
- ☐ Government-assisted sales

90. Why choose the current sales channels? [Single choice]

- ☐ Higher price
- ☐ Cash on delivery
- ☐ Desire to sell as soon as possible
- ☐ Recommended by a friend
- ☐ Recommended by brokers
- ☐ Lacking understanding of alternative sales channels
- ☐ Other

91. What are the sources of information for understanding sales channels? [Multiple choice]

- ☐ Broker intermediation

- ☐ Recommended by a friend
- ☐ Proprietary sales channels
- ☐ Government promotion
- ☐ Mass media
- ☐ Proactively sought by buyers

92. What is the cooperation model with brokers? [Single choice]

- ☐ Commission based on unit weight
- ☐ Lump-sum payment

(If Q91 includes the first option)

93. What is the unit price paid to brokers? [Open-ended]

(RMB per jin)

(If Q92 selected first option)

94. How much fee was paid to the broker? [Open-ended]

(RMB)

(If Q92 selected second option)

95. What is your level of satisfaction with the current economic benefits of the orchard? [Single choice]

- ☐ Very Dissatisfied

- Dissatisfied
- General
- Satisfied
- Very Satisfied

96. What is your level of satisfaction with the current management costs of the orchard? [Single choice]

- Very Dissatisfied
- Dissatisfied
- General
- Satisfied
- Very Satisfied